

Worksheet answers

1. Outline differences between the three trees in Figure 1.

The relationship between the five primates differs between each tree.

In trees B and C chimpanzees and humans are sister species, but in tree A they are on separate branches.

In tree A chimpanzee, gorilla and orang-utan are sister species but have a different relationship in trees B and C.

In tree A human is on a separate branch while in trees B and C human is sister species to chimpanzee.

The outgroup (least related to the other primates) is gibbon in two of the trees, but orang-utan in tree B.

2. Comment on **why** you think there are differences between the three trees.

Tree A was developed prior to technical advances in molecular biology (approx 1950) and computing power. Study of evolutionary biology has significantly advanced since then.

Tree B, based on 42 nucleotides, is significantly different from tree C, based on 1140 nucleotides, in relatedness of gibbons and orang-utans to other primates. More data provides greater resolution.

3. The most current understanding of evolutionary relationship between Hominoidea is that represented in tree C. Why do you think this evolutionary relationship is now accepted?

Molecular evidence, such as that demonstrated by tree C, provides strong evidence of relatedness of primates to each other. When molecular evidence is compared to other forms of evidence, such as physical characteristics, fossils and embryological structures, a more complete picture of primate relatedness has emerged. It's now well accepted that humans and chimpanzees share a common ancestor and that gorillas are our next closest relative.

4. The three evolutionary trees shown in Figure 4 look different from each other. Do the trees show the same relationships between the organisms? Identify tips, nodes and branches in tree 2.

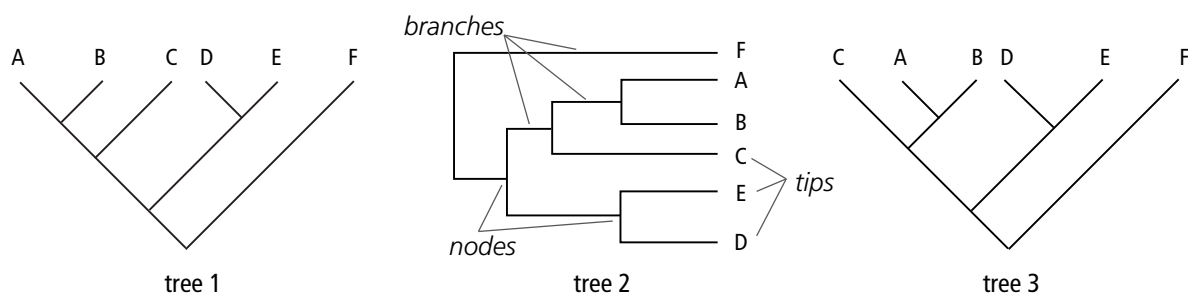


Figure 4: three evolutionary trees

The trees show the same relationships.

5. Figure 5 shows a full primate evolutionary tree. Mark the following on the tree:

- Label the most recent common ancestor (MRCA) of the Cercopithecinae (macaque) clade and Hominidae (great ape) clade with 'A'.
- Label the MRCA of Hominidae and Hylobatidae (gibbon) with 'B'.
- Label the MRCA of Cebidae (squirrel monkey) and Hylobatidae with 'C'.
- Which are most closely related to Hominidae: Colobinae or Cebidae? *Colobinae*
- Which are most closely related to Atelidae (howler monkey): Hylobatidae or Lemuridae (lemur)? *Hylobatidae (gibbon)*

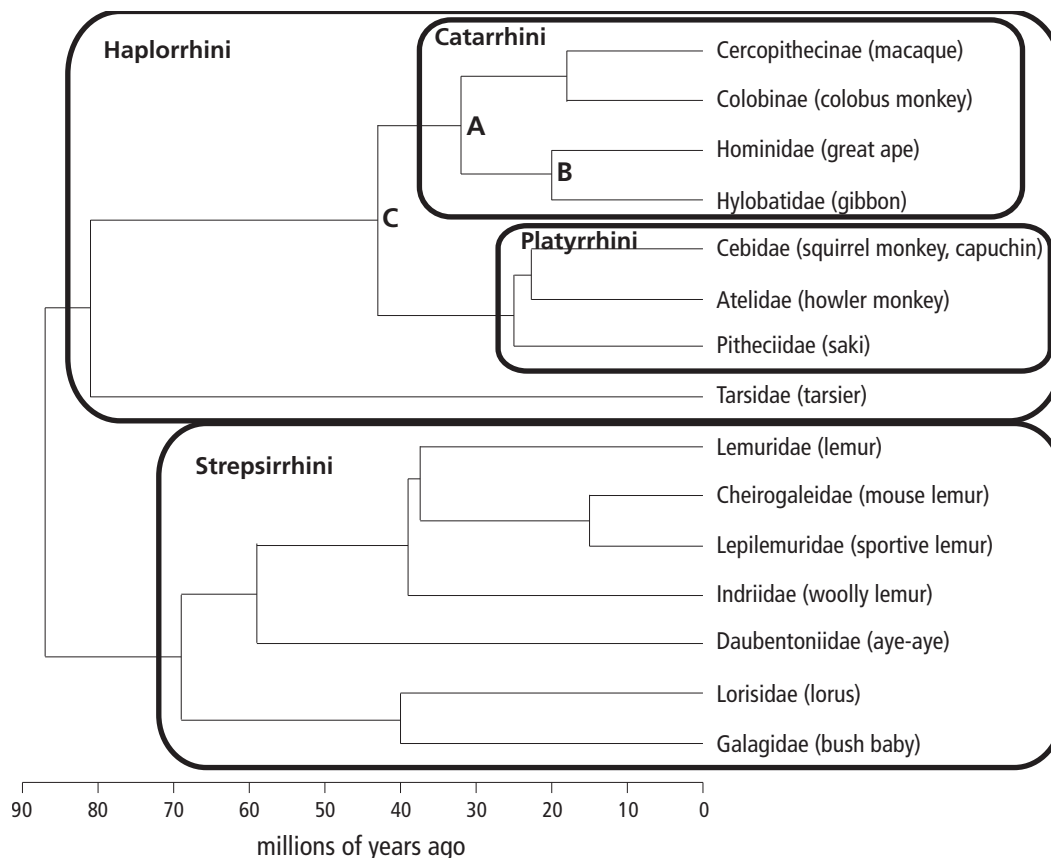


Figure 5: primate evolutionary tree

6. The order Primate forms a clade. Nested within this clade are other smaller clades that group primates according to shared characteristics. All primates share similarities, but some share particular features that others don't have. Circle the following four clades on Figure 5.

CLADE	MEMBERS	SHARED CHARACTERISTICS
Catarrhini	old world monkeys and apes	downward-facing nostrils, dry nose, large brain, sexual dimorphism
Strepsirrhini	prosimians	wet nose, protruding snout, tooth comb, upper lip attached to gum
Platyrrhini	new world monkeys	outward-facing nostrils, dry nose, opposable big toe, long tail, often prehensile
Haplorhini	old world monkeys, new world monkeys, tarsiers and apes	dry nose, larger brain, reduced snout, upper lip not attached to gum

7. Figure 6a shows an evolutionary tree for the Hominidae clade. In Figure 6b the same tree has been rearranged. Label the tree tips in Figure 6b so that evolutionary relationships between clade members remain the same.

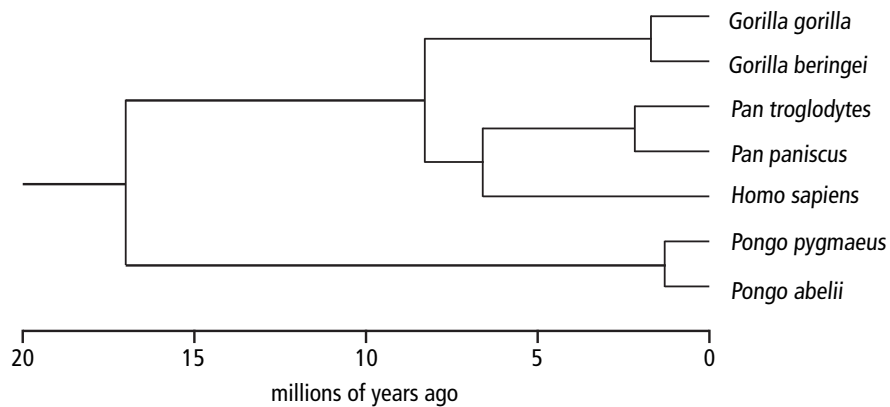


Figure 6a: Hominidae clade

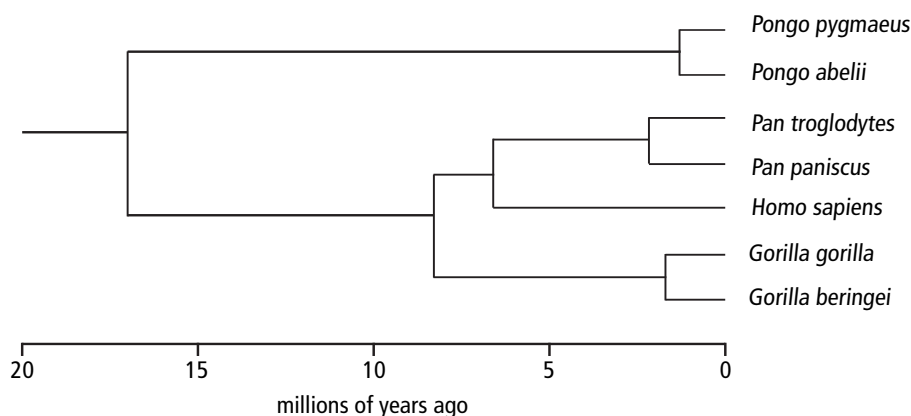


Figure 6b: Hominidae clade (rearranged)

8. Outline advantages and disadvantages of using cytochrome *b* to study evolutionary relationships, between species. Hint: Read the fact sheet, *What is molecular evidence?*

Cytochrome b is a housekeeping gene. Housekeeping genes are usually involved in some form of cellular maintenance or key process. As such, they're expressed at a constant level in most tissues. As these genes are usually highly conserved, differences in their sequences can be used in evolutionary studies.

Cytochrome b is mitochondrial DNA (mtDNA). It is been used in evolutionary research because, while there is a smaller amount of DNA in the mitochondria to manipulate, there is lots of it in a cell.

However, mtDNA does show a higher rate of genetic variation than nuclear DNA (nDNA). This may poorly reflect nDNA variation and if used in evolutionary studies may distort construction of phylogenies.

9. Consider the basic profile of the genes listed below in Table 2. Decide which gene you think would be a potential candidate gene to study evolutionary relationships. Give a reason for your decision.

There are two potential genes: EIF2B2 and ND4. Both of these are housekeeping genes and therefore will be expressed in high amounts in all tissue.

EVPL may not be expressed in all species being studied as it codes for a specialised protein.

EIF2B2 is nDNA meaning it will be more stable and therefore reflect DNA evolutionary trends more reliably.

ND4 is mtDNA meaning it will be in abundance and, with the higher mutation rate, potentially show more genetic variation.

GENE	nDNA	mtDNA	BIOLOGICAL FUNCTION
EVPL (envoplaken)	yes	no	protein coding: involved in development of knuckle pads
EIF2B2 (eukaryotic translation initiation factor 2B subunit 2β)	yes	no	protein coding: involved in the initiation and control of protein synthesis
ND4 (NADH dehydrogenase 4)	no	yes	protein coding: involved in energy production (mitochondria)

Table 2: potential candidate genes for evolutionary analysis

10. A common misconception within the wider community, when discussing evolution, is that humans evolved from chimpanzees. How would you explain what is really going on?

If humans evolved from chimpanzees there would be no chimpanzees. They would have all turned into humans! Chimpanzees are considered our closest relatives and, at some stage, estimated to be 6 million years ago, chimpanzees and humans shared a common ancestor.

11. In recent decades some countries, including the European Union, Japan, Australia and New Zealand, have imposed a ban on the use of Hominidae ('great apes') in research. As well as humans, this taxonomic family includes chimpanzee, bonobo, gorilla and orang-utan.

Other countries (Spain, Austria) have taken a step further and proposed that all Hominidae be granted legal rights. This would provide them the same basic protections afforded humans, including: the right to life; and freedom from exploitation and torture, and the right of consent in the participation in scientific research.

- a) How might molecular evidence for evolution contribute to future changes to animal welfare laws? What role might this evidence play in proposed legislation granting all Hominidae legal rights?

Proposed legislation to restrict, or ban, experimentation on all great apes commonly cites close evolutionary relationships between humans and the other great apes.

Along with genetic similarities, humans share social and cognitive similarities with great ape species. This raises ethical questions about the manner in which we treat them.

Animal welfare agencies cite this close genetic relationship between humans and other great ape species as strong support for the argument for granting legal rights to great apes.

Molecular evidence for evolution also strongly supports this argument, and has implications for further ethical discussions on the topic.

- b) Consider ethical concerns, and comment on advantages and disadvantages of using great apes in scientific research.

Advantages:

Great apes provide an excellent scientific model of the human condition: behaviourally, biologically and socially.

Great ape research allows us to study the origin and evolution of humans.

Close genetic links between humans and other great apes make these species, particularly chimpanzees, excellent models of human disease.

The use of great apes in research has led to important medical breakthroughs, including development of vaccines for human conditions, such as hepatitis A and B.

Research on great apes may help us develop treatments for complex human diseases, such as Parkinson's and Alzheimer's disease, and potentially alleviate suffering of many thousands of humans.

Future epidemics and disease outbreaks affecting humans may require experimentation on great apes to save human lives.

Disadvantages:

As humans may not be used in scientific research without consent, the close genetic relationship between them and other great ape species raises serious ethical questions about using those species in scientific research.

Great apes have complex social needs, most of which cannot be met under laboratory conditions.

Great apes exhibit many similar psychological capacities to humans. For example, animals used in experimental research have been documented to develop psychiatric conditions, such as post traumatic stress disorder.

Great apes do not always provide an accurate model of human disease. For instance, HIV acts differently in chimpanzees.

Maintaining great apes in captivity is expensive, and continued care after retirement from research requires ongoing costs.

Many great apes are endangered, and their use in experiments contradicts efforts to preserve their habitat and protect them in the wild.

The primate evolutionary tree in Figure 5 is based on Perelman, P., Johnson, W.E., Roos, C. et al., 2011. A molecular phylogeny of living primates. *Plos Genetics*, 7(3), 1-17. Accessed from <http://www.plosgenetics.org>

For further information on issues raised in question 12, visit the Great Ape Project, www.greatapeproject.org